Erica Bertolozzi

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Work Experience

Deloitte Consulting - New York, NY

Full-Stack Software Engineer II *Jun. 2025 - Present*

- Architected factory-based multi-agent system supporting 6 specialized AI security analysts, reducing agent instantiation code by 70% and enabling dynamic model switching;
- Engineered cloud security agent detecting 15+ threat types using MCP-based Athena queries, processing CloudTrail logs with <2s query turnaround;
- Integrated AWS Bedrock foundation models via custom provider abstraction, enabling cost-optimized model selection;
- Led the front-end development of seven generative AI-powered risk analysis tools, driving over 99% reduction in time required to generate cybersecurity control artifacts (from weeks to 10 minutes), demoed to agency stakeholders;
- Designed and implemented user interfaces end-to-end using Figma, collaborating with QA teams for rigorous testing while ensuring compliance with Deloitte branding standards and WCAG accessibility guidelines;
- Engineered backend integrations with Python/Flask and PostgreSQL deployed in Docker, enabling seamless data flow and supporting high-velocity feature development across all analysis tools;
- Developed full-stack React/Python resume automation tool with MongoDB data layer, Azure QA deployment pipeline and CI/CD integration, accelerating production go-live by 3 weeks.

Full-Stack Software Engineer I *Sept. 2022 – Jun. 2025*

- Enhanced security and operational efficiency by architecting and implementing role-based access control for order placement and inventory management, ensuring that manufacturing clients could operate within a guarded permission framework;
- Created and optimized real-time factory simulation dashboards using React, enabling manufacturing clients to visualize key performance indicators, identify bottlenecks, and proactively address operational inefficiencies, resulting in measurable reductions in plant downtime;
- Developed and deployed an interactive changeover time widget, equipping users with actionable insights that drove continuous process improvement and supported data-driven decision-making on the factory floor.

Business Technology Analyst *Sept. 2021 – Sept. 2022*

- Directed end-to-end user acceptance testing for a client-facing loan portal, ensuring alignment with business requirements. Drove successful go-live by validating user flows and system reliability in pre-launch.

Business Technology Summer Scholar *July 2020*

- Completed a virtual consulting program developing client engagement and problem-solving skills through a simulated project optimizing restaurant operations during the COVID-19 pandemic.

General Dynamics Mission Systems - Pittsfield, MA

Software Engineer Intern *May 2019 – Jul. 2019*

- Developed a high-performance simulation tool using C++ socket programming and multithreading, modeling critical system interactions for an early-stage defense project.
- Contributed to mission-critical software development under a U.S. Department of Defense security clearance, handling confidential information with precision and integrity.

Education

Rensselaer Polytechnic Institute (RPI), Bachelor's Degree in Computer Science May 2021

Skills and Certifications

Languages: JavaScript/TypeScript, Python, Java, .NET/C#, HTML

APIs and Databases: RESTful APIs, Flask, PostgreSQL, MongoDB

Technologies: git, Docker, Figma, Firebase, Claude Code, Google ADK, AWS Bedrock

Certifications: Google Cloud GenAI Leader, PCEP Certified Entry-Level Python Programmer, Google Cloud Digital Leader

Interests: Yoga, Weightlifting, Video Games, Concerts, Puzzles, Local Volunteering